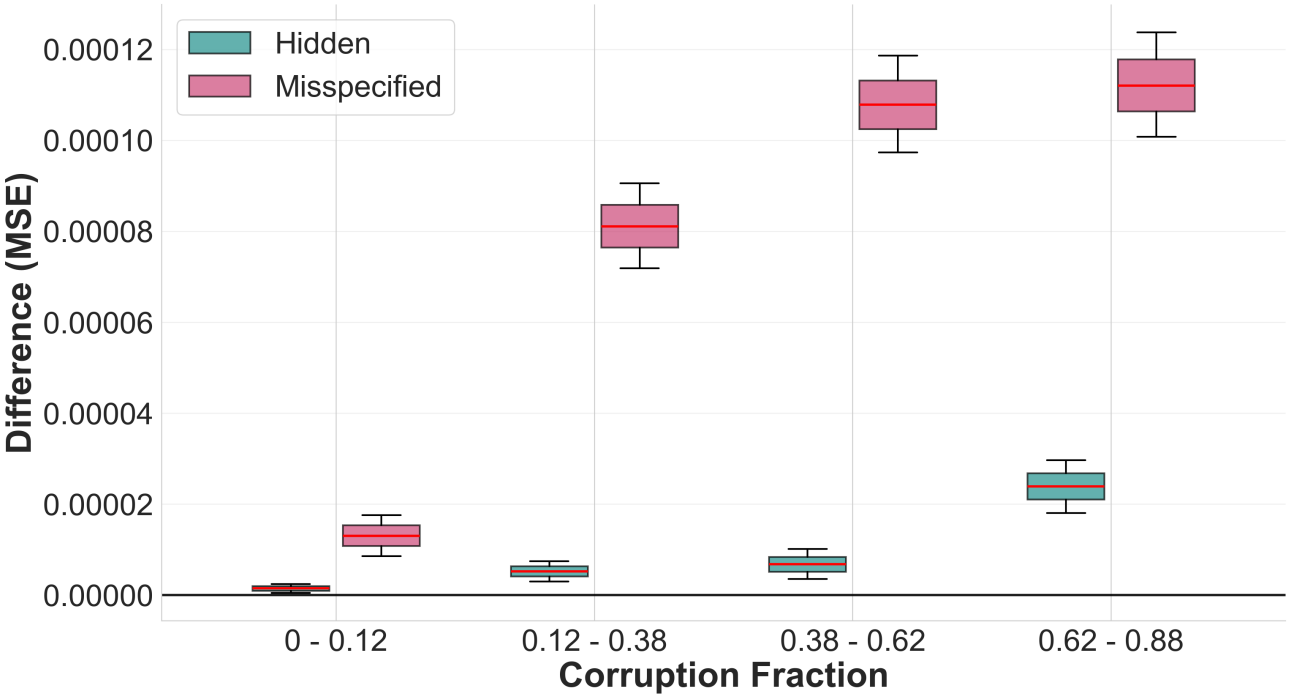

Figure 1. Difference between hiding and misspecifying edges in the partially known ancestral matrix evaluated on 100,000 samples from our complex mechanism prior. The baseline is a graph-conditioned model (GCN + SoftAttention) that takes as input the full ancestral matrix and we compare the performance of the same graph-conditioned model when corrupting fractions of the ancestral matrix. We either change a certain fraction of the entries to "hidden" by setting the selected entries $T_{i,j} \in \{-1, 1\}$ to $T'_{i,j} = 0$. Alternatively, we misspecified the same fraction of entries by flipping their sign $T'_{i,j} = -T_{i,j}$. We find that, while the performance of the model with increasingly hidden edges slowly degrades in performance relative to the baseline, the model with misspecified edges exhibits a substantially larger decrease. This further underscores the importance of being able to specify "unknown" for a specific causal relationship rather than having to settle with a fixed causal direction even though this might be incorrect.